

Moamen Moustafa Basyoni

R&D Engineer — Backend & Distributed Systems

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PROFESSIONAL SUMMARY

Backend engineer with roughly three years of production C#/.NET experience, building RESTful APIs and event-driven, distributed backend services. Currently replacing a legacy distributed-transaction design with Temporal Saga orchestration and building the API gateway and service-discovery layer around it — direct experience with the event-driven and distributed-systems patterns this role values. Also brings hands-on payments integration (Stripe, Paymob, webhook/HMAC verification), which overlaps with fintech-style correctness requirements even though the employer wasn't a fintech itself.

PROFESSIONAL EXPERIENCE

R&D Engineer — Backend, Distributed Systems & AI Engineering

Andalusia Health Business Solution (AHBS) · Alexandria, Egypt · Dec 2023 – Present

- Build and maintain RESTful APIs and services with **C#, ASP.NET Core (.NET 8/9/10)** and **Entity Framework Core** over **SQL Server** and PostgreSQL.
- Replaced a legacy MS DTC distributed-transaction workaround with **Temporal Saga orchestration**, improving 10+ inter-service interaction patterns — direct experience with distributed, event-driven correctness problems.
- Implemented **event-driven communication with RabbitMQ**, a **Consul**-based service-discovery layer, and an **Nginx API gateway** with parallel service versioning, cutting deployment downtime from 2 hours to 15 minutes (87.5%).
- Designed and built an **OIDC/OAuth2 identity server** (OpenIddict, JWT, scope/claim-based authorization).
- Modernized 5 core + 7 additional modules off legacy .NET Framework 4.5, applying Clean Architecture and modern coding standards.
- Test coverage with **xUnit**; a runtime business-rule engine benchmarked to a 45% performance improvement with BenchmarkDotNet.
- Migrated source control from a TFVC monolith to **Azure DevOps Git multi-repo**; established CI/CD pipelines across Dev, Staging and Pre-Live.

Technical Partner (equity) — Trastain

Cairo, Egypt (Remote) · Feb 2025 – Aug 2025

- Integrated **Stripe and Paymob** payment gateways for booking transactions, with webhook processing and **HMAC signature verification** — direct payments/fintech-adjacent integration experience.
- Handled transaction security for booking execution on a platform built from scratch.

Software Engineer — MYM (Make Your Miracle)

Giza, Egypt · Jun 2023 – Oct 2023

- Built backend systems for e-commerce and booking platforms, delivered end to end.

Freelance / Independent Client Work

- Integrated Stripe/Paymob payments with webhook/HMAC verification; delivered on both Azure and AWS depending on project needs.

TECHNICAL SKILLS

- **Languages:** C#, Python, SQL, TypeScript, JavaScript
- **.NET / Backend:** C#, .NET Core, .NET 6+/8/9/10, ASP.NET Core, Entity Framework Core, RESTful APIs, gRPC
- **Databases:** SQL Server, PostgreSQL
- **Distributed Systems & Messaging:** Temporal (Saga orchestration), RabbitMQ, MassTransit, gRPC, event-driven architecture
- **Architecture:** Clean Architecture, Microservices, Service-Oriented Architecture, Distributed Systems
- **Service Discovery & Gateway:** Consul, Nginx API Gateway, dynamic routing, load balancing
- **Identity & Security:** OpenID Connect, OAuth2, OpenIddict, JWT, scope/claim-based authorization
- **Payments & Integrations:** Stripe, Paymob, webhook processing, HMAC signature verification
- **Testing:** xUnit, BenchmarkDotNet, pytest
- **Cloud:** Azure (App Service, Blob Storage, SQL, DevOps) and AWS (EC2, S3) — basic-to-solid exposure on both
- **DevOps & Version Control:** Git, Azure DevOps Pipelines, GitHub Actions, CI/CD automation, Docker
- **AI-Assisted Development:** Claude Code, GitHub Copilot

EDUCATION

B.Sc. Computer Engineering — Behera Higher Institute (BHI), Alexandria, Egypt · 2023 Grade: B+ · GPA 3.2 / 4.0

PUBLICATION & LANGUAGES

- *IoT-Enabled E-Prescription Management and Dispensing Machine Monitoring* — IEEE, 2024
- Arabic (Native) · English (B2, Upper Intermediate)